A Method for the Analysis of Small Crop Fields in Sentinel-2 Dense Time Series

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***Abstract*— Satellite image time series (SITS), such as those by Sentinel-2 (S2) satellites, provides a large amount of information due to their combined temporal, spatial, and spectral resolutions. The high revisit frequency and spatial resolution of S2 result in: 1) increase in the probability of acquiring cloud-free images and 2) availability of detailed information for analyzing small objects. These characteristics are of interest in precision agriculture, where temporally dense SITS can benefit the understanding of crop behaviors. In the past, information about agricultural practices has been collected over large regions and focused on mixed/aggregated crops due to the poor tradeoff between the spatial and temporal resolutions. Products have been generated at low spatial resolution and daily basis or at high spatial resolution and weekly/monthly basis. They are meaningful for large agricultural fields, whereas they are limited when fields show a small average size. In this context, S2 characteristics allow for both high spatial and temporal resolution products. However, no existing automatic method effectively separates small fields from each other in an unsupervised way and deals with data irregularly sampled in time. Thus, this article presents a method suitable for the analysis of small crop fields in S2 dense SITS that accounts for S2 characteristics. The method fuses spatio-temporal information, analyzes data spatio-temporal evolution, and extracts relevant spatio-temporal information. The effectiveness of the proposed method was corroborated by experiments carried out on S2-SITS acquired over an area located in Barrax, Spain.**

***Index Terms*— Nonparametric regression, precision agriculture, satellite image time series (SITS), Sentinel-2, temporal evolution, vegetation phenology.**

I. Introduction

FOR decades, Earth observation (EO) satellites provide a unique way to observe our living planet from space. Due to the revisit orbit of the EO satellites, a huge amount of multitemporal images are now available in archives and continue to be acquired. The temporal, spatial, and spectral information in such satellite image time series (SITS) offers unique opportunities for monitoring and managing land cover dynamics from local to global spatial scales [1]. SITS provides temporally and spatially consistent spectral measurements of Earth surface that correlate directly with annual/interannual cycles of vegetation growth (i.e., phenology). Therefore, SITS offers a significant insight into the phenological evolution of vegetation at short and long terms, which is suitable for precision agriculture [1]–[3]. In this context, the opportunity to extract precise information about global agricultural systems is becoming more relevant as the food need increases. Nevertheless, little high spatial/temporal resolution cropland maps from remote sensing are available in the literature, which provides [4]–[7]: 1) areal extent; 2) cropping intensity; and 3) state at crop field level. Extracting such information from remote sensing data is challenging, especially when considering intensively cultivated areas with small-size crop fields and frequent crop rotation (such as Europe). To accurately model agricultural areas with those characteristics, SITS should satisfy certain characteristics. First, SITS should be continuous and regular for crop identification and phenological parameters derivation. State-of-the-art (SoA) methods are able to generate continuous SITS, but they require regular acquisitions in time [2], [8]–[11] and *a priori* knowledge (e.g., the rotation practices) of the agricultural area. Second, SITS should be temporally dense in order to properly sample the fast variations of the crops (especially herbaceous ones), temporal signature that usually shows short phenological cycles with fast maturation and fallow stages [12]–[15]. Finally, SITS should show a spatial resolution adequate to the size of crop fields (parcels). This is a recurrent issue [5], [6], [15] since the field size is usually small with respect to the sensor spatial resolution (e.g., MODIS and Landsat) [5], [16]. The poor ratio between the crop field size and the sensor spatial resolution makes precision agriculture at field level a complex task [15]. To overcome the issue, SoA methods usually perform analysis at single pixel level; however, this requires the selection of meaningful and representative pixels by means of a ground truth that is usually not available. Summarizing, producing detailed cropland maps requires SITS with high temporal and spatial resolutions and regular time spacing. To date, data from a single satellite sensor were not capable of simultaneously providing such characteristics. Some of them (like Landsat) provide high spatial resolution

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(30 m) but poor temporal one (16 days), whereas others (such as MODIS) show high temporal resolution (1–2 days) but coarse spatial one (0.25–1 km) [17]. Therefore, past generation data do not provide the capability to capture the fast changing crop field behavior at the parcel level [15], [17]. Instead, nowadays, the S2 constellation of the European Space Agency (ESA) Copernicus Program provides both high temporal and spatial resolutions SITS. S2 constellation is made up of twin satellites: S2A (launched in 2015) and S2B (launched in 2017). Jointly, they fully cover Earth every five days (at the Equator). Since S2 orbit is polar, the revisit period increases as we move from the Equator to the Northern Hemisphere as a result of overlapping orbits [5], [18]–[20]. S2 images have a spatial resolution from 10 to 60 m in 13 spectral bands. Together with the high temporal resolution and the wide swath of S2 (290 km), overlapping orbits increase the number of acquisitions over a specific geographical area (which satisfies the second requirement) and thus in a higher probability of acquiring free cloud images. Despite that S2-SITS is still irregular in time, they make it possible to generate products with a better compromise in terms of temporal and spatial resolutions [14], [15], [21]. The latter is high enough to allow for an analysis at single crop field level even with small/dense fields. To this end, the challenges of identifying single crop fields and dealing with irregularly acquired data should be tackled.

S2 data have the potential to allow for separation among crop fields. This separation opens to the possibility of performing temporal analysis at a single field level, where the common practices up to date were to either perform aggregate field analysis or to analyze only single meaningful pixels. This was due to the lack of reliable cadastral maps and to the poor performance of solutions based on crop field boundary detection by segmentation. The main limitation of the latter one is the use of one or few acquisition dates [22]. This choice does not allow to model the high temporal variability of dense and intensively cultivated areas showing several types of crops at different phenological stages. Crops behaviors are captured at some stages of their vegetative cycle only resulting in missegmentation of fields, especially the fallow ones. Some methods, such as the ones presented in [23] and [24] and the ones summarized in [25], address the problem by means of a hierarchical segmentation (mean shift, multiresolution, multithreshold, and watershed among others). Such methods require several parameters to be tuned and suffer from oversegmentation, making the results unreliable. Other methods exist [26], which consider both segmentation-based mean shift and supervised classifiers to separate among crop and noncrop areas. However, they also suffer from oversegmentation and the ground-truth data are often not available. Thus, the relevance of a crop field map that accurately separates each field (in an automatic way) becomes clear. This can be achieved by properly modeling the phenological cycle. In this context, dense SITS represents a precious source of information from both phenological and crop field boundaries perspectives [15], [27]–[29]. Dense SITS not only helps for crop fields' detection but also to decrease the impact of possible missing/corrupted observations due to atmospheric conditions [5], [6], [30]. Therefore, in this article, we take advantage of the crop phenological behaviors and the spectral and spatial characteristics of dense S2-SITS to build a crop field map that considers the multitemporal information.

Once crop fields have been separated and their boundaries detected, fields' behaviors can be analyzed at field level in terms of phenological parameters' estimation from the crop dynamic. To track the month-to-month dynamics, noise-free and/or continuous sampled data are required. However, S2-SITS is nonuniform and irregular in time (nonperiodic). This is due to the common atmospheric and geometric distortions [8], [31]. Therefore, steps are required for denoising and gap filling [31]. SoA methods for missing or noisy data management mostly rely on smoothing and function-fitting algorithms [2], [9], [11], [13], [32]–[34]. Those methods assume that data are regularly sampled in time (except for [34]) and the temporal signature shows a harmonic behavior with one (or maximum two) cropping cycles. Methods, such as the one presented in [34], also require prior information on the first part of the temporal information in order to determine a shape prior. In other words, with three-year information, it can only estimate reliable information on the central one-year of the SITS. These algorithms reduce the amount of missing data while increasing the capability of extracting phenological parameters [35]. However, the abovementioned assumptions hold for crop analyses conducted on aggregated fields and/or in low intensive agriculture areas, where crops tend to show one harmonic cycle per year, whereas they fail when focusing on single crop fields in intensively cultivated areas and showing more harmonics per year. Thus, when applied to S2-SITS, they tend to result in poor estimates of phenological parameters, such as the number of cropping cycles, length of season, middle of season, and maximum value. Other methods derive the phenological parameters independently of the SITS smoothing, but they require user manually defined thresholds [9], [11]. Therefore, there is a need to develop and/or adapt SoA algorithms/methods to the analysis of temporally dense and irregularly sampled S2-SITS in the context of precision agriculture and reduce the need of supervision.

This article presents a method for the analysis of small crop fields in S2 temporally dense time series. The method automatically identifies and separates crop fields by considering their phenological behavior over a time range, reconstructs continuous and regular sampled S2-SITS using an adaptive nonparametric regression model, and extracts spatio-temporal information about crop fields. The proposed method is automatic, and products can be updated when new images are available. It deals with irregularly sampled SITS agricultural areas with small and intensive cycle crop field.

The remainder of this article is structured as follows. Section II introduces the proposed method for studying the spatial and temporal behaviors of small crop fields in S2-SITS. Section III describes the data set used to validate the proposed method, presents the results for its different stages, and compares the results with SoA methods. Finally, Section IV draws the conclusions and illustrates future works.

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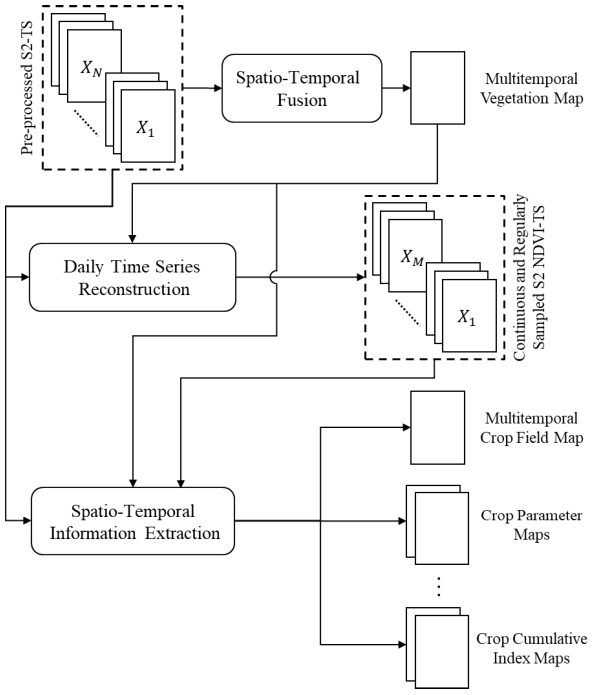


Fig. 1. Block scheme of the proposed method.

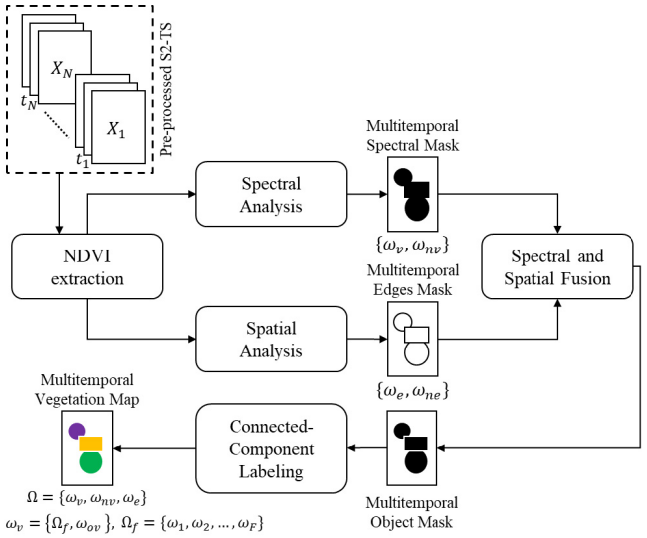


Fig. 2. Block scheme of the proposed approach to spatio-temporal fusion.

II. Analysis of Small Crop Fields in Sentinel-2 Dense Time Series

The proposed method models the spatial behaviors and the temporal evolution of crop fields in high spatial and temporal resolution S2-SITS to generate a multitemporal crop field map. The multitemporal concept comes from the fact that all the images from the S2-SITS are used to this end. Each crop field is analyzed independently for crop-field-level product generation (e.g., phenological parameters and cumulative indexes). Here, the products are generated considering vegetation indices. Since we are dealing with agricultural application, normalized difference vegetation index (NDVI) is used [22], [36], [37]. Fig. 1 shows the block scheme of the proposed method that is based on three main steps: 1) spatio-temporal fusion; 2) daily time series reconstruction; and 3) extraction of relevant spatio-temporal information. First, a multitemporal vegetation map is built by fusing spectral, spatial, and temporal (NDVI) information. Second, the multitemporal vegetation map is used as a mask in order to build regular and continuous NDVI time series at the level of single object/crop field. Finally, the multitemporal vegetation map and the daily NDVI-TS are used to extract spatio-temporal information. Among the possible products relevant to agricultural practices, here, we focus on phenological parameters and cumulative indices. However, other products can be designed at need, according to experts' criteria.

Let SITS $= \{X_1, X_2, X_3, \ldots, X_N\}$ be a temporally dense time series of S2 images acquired over the same geographical area characterized by agricultural activities in a period $[t_1, t_N]$. Let us assume that images are acquired with nonuniform time sampling (irregular SITS). Let Ω_F ($f \in [1, F]$) be the set of F existing fields in SITS. Ω_F includes both cultivated and fallow crop fields. Let us finally assume that crop fields in X_n ($n \in [1, N]$), are cultivated with different types of crops, and thus, they show different phenological cycles and space and time varying behaviors and that intensive crop rotation is practiced in the area (i.e., fields are cultivated with different crops over the seasons).

A. Spatio-Temporal Fusion

The spatio-temporal fusion detects and separates the possible crop fields in SITS by considering both S2-SITS spatial resolution and the temporal phenological evolution of vegetation. Fig. 2 shows these steps applied to an NDVI time series. The NDVI S2-SITS spectral and spatial information is analyzed to generate a multitemporal object mask (where spatially contiguous pixels may correspond to a single crop field). Finally, each object is separated by connected-component labeling, leading to a multitemporal vegetation map. Depending on the study area, these objects may correspond only to crop fields (Ω_F) or to other kinds of vegetation (ω_{ov}), such as forest or wetlands (that in turn show different temporal evolution in SITS). Thus, the multitemporal vegetation map includes three classes $\{\omega_v, \omega_{\text{nv}}, \omega_e\}$. $\omega_v = \{\Omega_F, \omega_{\text{ov}}\}$ is the class of objects such that vegetation is detected for at least one $t_i \in [t_1, t_N]$, $\{\Omega_F = \{\omega_1, \omega_2, \ldots, \omega_F\}$ is the class of objects identified as crop fields (both cultivated and fallow ones), and ω_{ov} represents other vegetation, such as forest or grass. ω_{nv} represents all the objects that never showed vegetation, such as buildings, roads, and water bodies. ω_e represents the crop field edges. The most common approach to isolate ω_v in the multitemporal vegetation map is to use the information about "agricultural areas" from an available land cover map [38], [39]. The problem of this strategy is that available land cover maps are usually representative of one (or few) acquisition date(s) only. Therefore, they do not fully capture the multitemporal variability of SITS.

1) NDVI Extraction: One of the key elements of spatio-temporal fusion performance in separating $\{\omega_v, \omega_{\text{nv}}, \omega_e\}$ is the

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choice of adequate features. In the literature, original spectral bands, radiometric indices, and so on are used, showing the capability to highlight different properties of vegetation behaviors [40], [41]. These radiometric indices exploit the spectral information acquired by the sensor bands and analyze their physical response to measure biomass or vigor of vegetation. Here, NDVI is used since it is one of the most effective recognized indices [22], [36], [37]. The NDVI equation is given in the following, where $B8a$ and $B4$ correspond to the near-infrared (NIR) and red bands, respectively, to highlighting the presence of vegetation:

$$\text{NDVI} = \frac{B8a - B4}{B8a + B4}. \tag{1}$$

NDVI generally assumes large positive values when healthy green vegetation exists, due to both absorption and reflection of most visible and NIR light. Extreme negative values are observed when unhealthy or sparse vegetation and water are present. Finally, NDVI is close to zero for bare soil [42]. The method extracts NDVI for each X_n (X_{NDVI_n}) in SITS and generates an NDVI-SITS with values in the range $[-1, 1]$. It is worth noting that similar considerations apply to any feature, and however, from now on, we will refer to NDVI.

2) Spectral Analysis: According to the NDVI characteristics, a threshold $0 < T < 1$ can be applied, which separates ω_v from $\omega_{\text{nv}}, \omega_e$ in each X_{NDVI_n} ($n \in [1, N]$). The value of T is expected to remain the same along different agricultural areas. Therefore, it is rather easy to set it up by fast trial and error. This information is used to conduct a multitemporal analysis and identify all objects being detected as vegetated at least once in $[t_1, t_N]$. The multitemporal variability of crop fields is coded into a multitemporal binary spectral mask $X_{\text{Spectral}_{\text{mask}}}$ that is defined as

$$X_{\text{Spectral}_{\text{mask}}} = \begin{cases} \omega_v, & \text{if } \left(\sum_{n=1}^{n=N} (X_{\text{NDVI}_n} > T) \right) \geq 1 \\ \omega_{\text{nv}}, & \text{otherwise.} \end{cases} \tag{2}$$

$X_{\text{Spectral}_{\text{mask}}}$ contains all the objects with vegetation (ω_v) and those with nonvegetation (ω_{nv}) along time variable, and ω_v includes crop field (Ω_F) and other vegetation (ω_{ov}) as well. However, $X_{\text{Spectral}_{\text{mask}}}$ does not account for spatial information. Thus, neighboring fields showing the same multitemporal NDVI behaviors are merged together. Neither does $X_{\text{Spectral}_{\text{mask}}}$ provide spectral distinction with other vegetation classes, such as forest or grass. To separate among different types of vegetation, further strategies are required. $X_{\text{Spectral}_{\text{mask}}}$ can be dynamically updated as a new acquisition X_{N+1} comes in.

3) Spatial Analysis: To integrate the process, the spatial information available in NDVI-SITS, let us recall that crop types showing different phenological patterns differ in the temporal dynamic behavior of NDVI. While each crop type will become strongest when close to crop maturity (i.e., the Middle of Season instant—$t_{\text{MoS}} \in [t_1, t_n]$), the timing of MoS will vary, resulting in detectable edges (X_{edges_n}) between the neighboring field crop objects. The larger this difference, the stronger the edge, and the higher the probability of separating neighboring crop field objects. X_{edges_n} ($n = 1, \ldots, N$) are jointly used to separate edges (ω_e) from no edges (ω_{ne}) over time. To ensure the selection of reliable edges, only strong edges are preserved. These are the ones that appear at least twice in X_{edges_n} ($n = 1, \ldots, N$) (3). Therefore, $X_{\text{Edges}_{\text{mask}}}$ cumulates multitemporal boundaries of objects in ω_v

$$X_{\text{Edges}_{\text{mask}}} = \begin{cases} \omega_e, & \text{if } \left(\sum_{n=1}^{n=N} X_{\text{edges}_n} \right) > 1 \\ \omega_{\text{ne}}, & \text{otherwise.} \end{cases} \tag{3}$$

A low probability exists that some areas corresponding to other vegetation (ω_{ov}) generate strong edges due to their proximity to crop fields and the sensor spatial resolution. Without loss of generality, the threshold in (3) can be slightly increased to reduce spurious strong edges generated by temporary events (e.g., the presence of a cloud). As $X_{\text{Spectral}_{\text{mask}}}$, $X_{\text{Edges}_{\text{mask}}}$ can be dynamically updated as a new acquisition X_{N+1} comes in. Among the others, Laplacian and Sobel operators are tested here to detect edges [43].

4) Spectral and Spatial Fusion: By combining the $X_{\text{Spectral}_{\text{mask}}}$ and $X_{\text{Edges}_{\text{mask}}}$ as in (4), all vegetated areas in the NDVI-SITS are identified and separated. Thus, X_{mask} is a binary mask where no-vegetation (ω_{nv}) and the crop field borders (ω_e) are separated from vegetation (ω_v). The borders in $X_{\text{Edges}_{\text{mask}}}$ are removed from further analysis since they tend to be mixed pixels

$$X_{\text{mask}} = X_{\text{Spectral}_{\text{mask}}} \cdot X_{\text{Edges}_{\text{mask}}}. \tag{4}$$

5) Connected-Component Labeling: To group spatially contiguous pixels in X_{mask}, automatic detection and numbering of isolated, yet connected, regions in the binary image is applied. Here, the well-known connected-component labeling method is used [44]. Pixel membership to a given region is evaluated by considering a four- or eight-connected neighborhood. The labeling results in a multitemporal vegetation map $X_{\text{veg}_{\text{map}}}$ with classes $\omega_v, \overline{\Omega_F}, \overline{\Omega_F} = \{\omega_{\text{nv}}, \omega_e\}$. From $X_{\text{veg}_{\text{map}}}$, we proceed to the single object temporal analysis.

B. Daily Time Series Reconstruction

To study the spatio-temporal evolution of crop fields, the presence of the vegetation cycles among different crops is usually analyzed [6], [9]. To this end, continuous and regularly spaced SITS are needed. Smoothing methods for NDVI-SITS available in the literature traditionally make two assumptions: 1) a temporal signature shows one (maximum two) t_{MoS} in a year and 2) the behavior of maturation cycles is close to be harmonic. Consequently, temporal signatures are usually reconstructed by harmonic oscillators and/or Fourier series [3], [34], [45] with an *a priori* user-defined number of peaks (also known as cropping cycles). However, modern agricultural practices push toward intensive cultivations where crops show several maturation cycles per year and multiple harmonics induced by rotation [46]–[48]. Furthermore, as illustrated earlier, SITS tends to be irregularly sampled in time. In this context, both harmonic oscillators and Fourier series-based methods fail [3], [10], [45]. Thus, rather than using one predefined model for each single pixel, we propose

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to define an *ad hoc* nonparametric and adaptive regression model for each single object in $X_{\text{veg}_{\text{map}}}$, by means of a multilayer perceptron neural network (MLP-NN). The approach is automatic and able to deal with irregularly sampled images. One of the many problems with SoA methods is the proper reconstruction of the information at the beginning and end of the SITS where discontinuity occurs, which is usually underestimated/overestimated. This is traditionally solved by: 1) including extra images acquired before and after the considered period or 2) by replacing the missing acquisitions by replicating the first and last images available in the SITS [9], [49]. The proposed approach is developed without such constraint by imputing the existing data with an adaptive nonparametric regression (by means of MLP-NN). Three main steps are followed: 1) definition and extraction of NDVI-SITS sets; 2) NDVI-SITS data imputation by upper envelope and withdrawn strategy; and 3) adaptive nonparametric regression of augmented NDVI-SITS.

1) Definition and Extraction of NDVI-SITS Sets: In order to perform the TS reconstruction, a set of NDVI training samples (NDVI_{tr}) is required. SoA methods have been limited by the low spatial resolution of sensors, resulting in lack of single object analysis. Thus, they do not select training samples, but rather use every single pixel in an image (including no-crop ones) to derive a regular TS. Then, they manually select the areas or pixels of interest (where the user is sure to find a crop field) to perform further analysis. The proposed approach accounts for the specific behaviors of a single crop by extracting the training samples at object level from $X_{\text{veg}_{\text{map}}}$. To mitigate noisy components, we define NDVI-SITS training sets (NDVI_{tr}) where training samples correspond to the mean NDVI value of objects computed date by date. Still, issues remain due to the irregularity of SITS.

2) NDVI-SITS Data Imputation by Upper-Envelope and Withdrawn Strategy: Irregular temporal data limits the accuracy of the estimation of phenological parameters. This is because estimation methods assume equally spaced regular-shaped (bell-like) data in the time domain. Thus, before applying the proposed regression step, preliminary symmetrical and uniformly sampled NDVI-SITS need to be generated. Usually, this is achieved by a linear combination of nearby values in the SITS in a window [9]. We propose an alternative approach, where a smoothed SITS is simulated generating daily acquisitions by data augmentation of the NDVI-SITS upper envelope with a withdrawn strategy. The steps of this strategy are illustrated by the example shown in Fig. 3.

1) For each NDVI-SITS set (NDVI_{tr}) defined in Section II-B1, select the samples that are above a fallow threshold (defined by trial and error as NDVI = 0.4). This threshold identifies when a given crop field can be considered as vegetated [green circles in Fig. 3(b)].
2) Calculate the local maxima (as the points with zero first derivative and negative second derivative) of the selected samples [purple circles in Fig. 3(c)] and withdrawn the remaining ones [gray samples in Fig. 3(c)] from NDVI_{tr}. This leads to the upper envelope of the data.
3) Use the samples below the fallow threshold [blue samples in Fig. 3(c)] and the local maxima from the previous step [purple samples in Fig. 3(c)] for data imputation by means of a piecewise cubic Hermite interpolating polynomial [PCHIP; orange line in Fig. 3(d)]. The selection of PCHIP over other interpolation methods is justified by its characteristic to preserve the shape of the data and respect monotonicity. The combination of blue and purple samples is defined as the upper envelope set.
4) Subtract the imputed data from NDVI_{tr}. Reinsert the withdrawn samples with a difference greater than zero to the upper envelope set [see the sample in the green circle in Fig. 3(e)]. This step allows to better follow the shape of the original data.
5) Impute the updated upper envelope set by means of PCHIP [purple line in Fig. 3(f)].

3) Adaptive Nonparametric Regression of Augmented NDVI-SITS: An MLP-NN with a single hidden layer is employed, which dynamically adapts to each object in ω_v. The design of object-dedicated MLP-NN allows to capture the specific temporal behavior of each crop field and to accurately follow the possible different levels of crop behavior complexity that cannot be modeled by a unique global model instead. This is achieved by automatically selecting the optimal number of neurons (n_{HL_i}) in the hidden layer. The selection of n_{HL_i} for each MLP-NN is conducted crop field by crop field by minimizing the mean square error (MSE) [50]. A given $n_{\text{HL}_i} = n_{\text{HL}_1}$ (lower bound) is first established by the user to train the MLP-NN. The MLP-NN is trained with an increasing number of neurons until MSE becomes lower than a threshold (T_{MSE}) or the number of iterations reaches a predefined upper bound (5). In the latter case, the iterative procedure stops with a warning of nonconvergence. The proposed MLP-NN is trained by using updated upper envelope PCHIP NDVI-SITS obtained from the previous step as target. The upper envelope PCHIP guarantees the preservation of the shape/behavior of the cropping cycle. However, this mathematical approach tends to overfit the data as signature complexity increases. The smoothed NDVI before (purple line) and after (green line) applying the proposed MLP-NN to the example shown in Fig. 3 are presented in Fig. 4

$$n_{\text{HL}_i} = \begin{cases} n_{\text{HL}_1} \\ n_{\text{HL}_{i-1}} + 1, & \text{if MSE} > T_{\text{MSE}} \text{ and } i < I. \end{cases} \tag{5}$$

C. Spatio-temporal Information Extraction

In the last step, we exploit the spectral, spatial, and temporal characteristics of S2-SITS for information extraction and complex products generation. Here, we focus on three options, but several others can be afforded as follows: 1) characterization and segregation of ω_{ov} and generation of the multitemporal crop field map ($X_{\text{crop}_{\text{map}}}$); 2) estimation of phenological parameters and generation of phenological maps at crop level; and 3) definition and analysis of crop-field-based cumulative indexes. To this aim, we use both $X_{\text{veg}_{\text{map}}}$ and the continuous and regularly sample NDVI-SITS.

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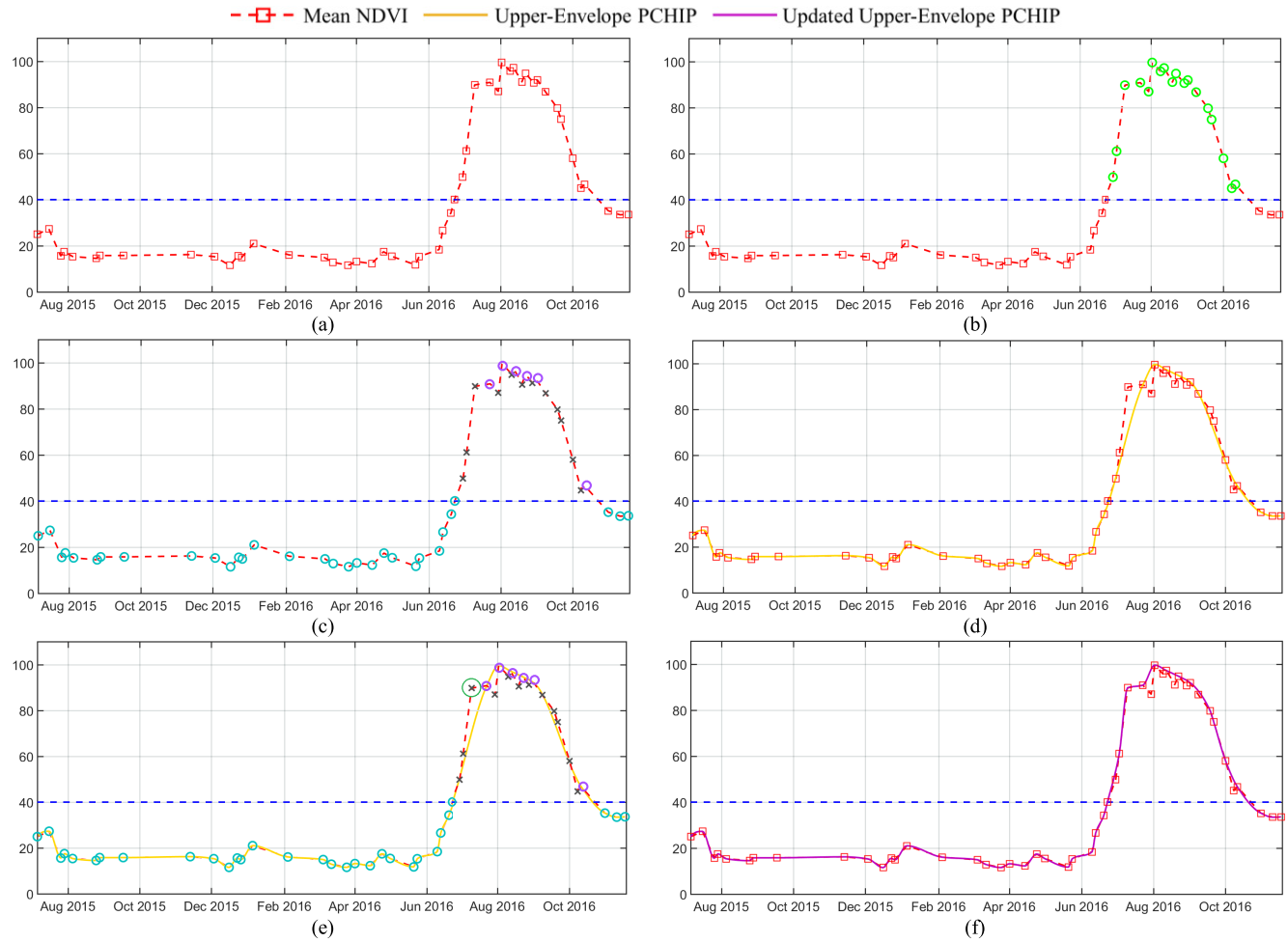


Fig. 3. NDVI-SITS set ($NDVI_{tr}$) of a toy example. (a) Mean NDVI. (b) Samples above the fallow threshold: green circles. (c) Upper envelope set: blue and purple circles. (d) Imputed upper envelope set: orange line. (e) Updated upper envelope set: blue, purple, and green circles. (f) Imputed updated upper envelope set: purple line.

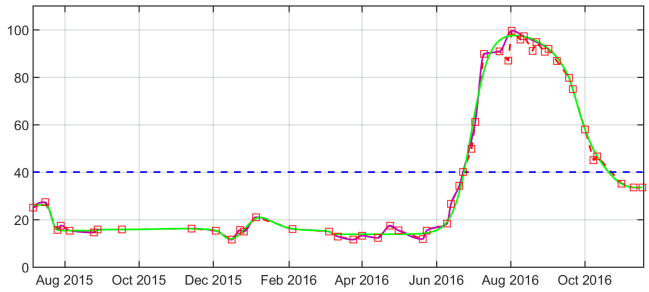


Fig. 4. Mean NDVI-SITS (red dotted line) and smoothed NDVI by imputed updated upper envelope set (purple line) and proposed MLP-NN (green line).

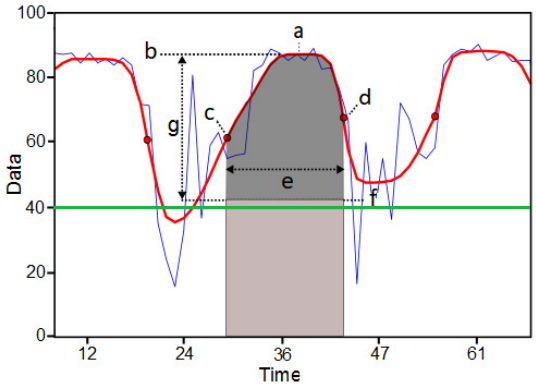


Fig. 5. Phenological parameters. (a) Time of MoS. (b) Maximum value. (c) BoS. (d) EoS. (e) LoS. (f) BV. (g) Amplitude (modified from [9]).

1) Phenological Parameter Maps: Events, such as budburst, leaf-out, or flowering, are of great relevance in precision agriculture. Their knowledge improves the understanding of crop evolution and thus supports crop management practices. Those events are related to well-known phenological parameters that can be derived from the NDVI-SITS. Fig. 5 locates those instants on the temporal signature: a) beginning of season (BoS); b) end of season (EoS); c) length of season (LoS); d) base value (BV); e) middle of season (MoS); f) maximum value (MV); g) amplitude; and h) number of cropping cycles. Available methods in the literature estimate them in a nonautomatic or semiautomatic manner [49], [51]. To allow for an automatic unsupervised estimation, here, we relate

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the phenological parameters to analytical behaviors of the temporal signature. In detail, we define: 1) MoS as the time of a local maximum; 2) MV corresponds to the maximum value of each MoS; 3) BoS as the descendent point of inflexion of the second derivative; 4) EoS as the ascendant point of inflexion of the second derivative; 5) LoS as the difference between EoS 2) and BoS 1); 6) BV as the lowest local minima on the left or the right side of a local maximum; 7) amplitude as the difference between MV and BV; and 8) number of cropping cycles as the number of detected MoS in the considered period. According to the abovementioned definitions, the automatic algorithm computes the first and second derivatives of the NDVI-SITS for each object in $X_{\text{veg}_{\text{map}}}$ and locates local maxima, minima, and inflection points. Finally, it derives the phenological parameters. The reliability of this procedure benefits from the improvements introduced in the NDVI-SITS reconstruction step. Other methods exist in the literature [52], [53], which exploits the first and second derivatives of a reconstructed NDVI-SITS, but they do not deal with more than one cropping cycle per year.

Knowing the phenological parameters, crop fields (Ω_F) can be distinguished from other types of vegetation (ω_{ov}—grass and forest, where ω_{ov} belongs to $\overline{\Omega_F}$) and further separated between the cultivated and fallow crop fields in $X_{\text{crop}_{\text{map}}}$. Cultivated areas show a bell-shaped NDVI behavior in the range [0, 1]. Fallow crop fields show NDVI-SITS with values smaller than the fallow threshold (NDVI < 0.4, green line in Fig. 5). The same fallow threshold applies for all types of crop, except for vineyards or orchard ones, where the percentage of bare soil tends to generate lower NDVI values. If BoS, EoS, and MoS fall below the fallow threshold, the number of cropping cycles is set to zero and no further analysis is conducted. Other vegetation, such as grass and forest, may show a bell-shaped NDVI-SITS profile such as the crop ones, but LoS is longer than that of crops (around six months). Thus, objects in ω_v with an LoS > 6 months are assigned to $\overline{\Omega_F}$. By means of these rules, the multitemporal crop field map ($X_{\text{crop}_{\text{map}}}$) with cultivated and fallow crop fields is generated. Phenological parameters can be used to generate phenological maps or combined to derive more complex products, such as winter/summer crop maps.

2) Cumulative Radiometric Indices Analysis: Among the complex products, we propose here cumulative indices, which are computed from the entire S2-SITS and are calculated at pixel or object level by summing over time ($n = 1, \ldots, N$) the instances of the radiometric index being above a threshold T (6)

$$X_D = \sum_{n=1}^{N} (X_{\text{Radiometric Index}_n} > T). \tag{6}$$

The accumulated radiometric index X_D provides a high-level representation about how crop fields evolved over time. The meaning of the cumulative index changes according to the considered radiometric index and the threshold (T) applied over it. As an example, let us consider the cumulated NDVI. Since healthy vegetation shows a response ranging in [0, 1], and setting $T = 0.5$, we define high cumulative NDVI values for fields with regular and healthy vegetation, and lower values may be associated with fields being less vegetated and/or stressed. Similar considerations hold for the normalized difference water index (NDWI), which is used to understand water content in crop fields [54]. The NDWI is calculated in a similar way to the NDVI [see (1)] but using the short-wave infrared (SWIR-B12) band instead of the NIR. Accumulated NDWI provides information about adequate water supply or about possible stress or lack of irrigation. Crop fields showing anomalies in cumulative indices can be analyzed in detail by considering the whole temporal signature rather than the cumulative one.

TABLE I
SENTINEL-2 SPECTRAL AND SPATIAL CHARACTERISTICS [57]

Band number	Central wavelength (nm)	Bandwidth (nm)	Spatial resolution(m)
1	442.7	21	60
2	492.4	66	10
3	559.8	36	10
4	664.6	31	10
5	704.1	15	20
6	740.5	15	20
7	782.8	20	20
8	832.8	106	10
8a	864.7	21	20
9	945.1	20	60
10	1373.5	31	60
11	1613.7	91	20
12	2202.4	175	20

III. EXPERIMENTAL RESULTS

A. Data Set Description, Preprocessing, and Ground Truth

In order to validate the proposed method, an SITS was built by considering the images acquired by S2A over the area of Barrax in the Southeast part of Spain. Barrax fits to tile T30SWJ that is observed by two overlapping orbits: orbit 094 and orbit 051. Therefore, we expect an image every 3–7 days. This increases the probability of having cloud-free images and guarantees denser SITS for agricultural analysis. A total of 76 images were considered, acquired from July 6, 2015, until November 30, 2016 (by the two orbits). Before applying any step, it is important to preprocess the S2-SITS. During preprocessing, each S2 tile was atmospherically corrected (i.e., converted from L1C to L2A level) by using the sen2cor tool provided by ESA [55]. The six bands at 20 m (see Table I) are upscaled to 10 m by means of an optimized high-pass filter (HPF) [56]. Band 4 is used as the panchromatic band for bands 5, 11, and 12, whereas band 8 is used for bands 6, 7, and 8A. The three 60-m bands are not upscaled since they do not provide relevant information about vegetation. Thus, here, they are used for atmospheric correction and cloud screening by sen2cor, only.

In our experiments, images with high percentage (greater than 70%) of irrelevant pixels are removed. Irrelevant pixels are the ones marked as cloud shadow, cloud low probability, cloud medium probability, cloud high probability, and thin cirrus according to the scene classification (SC) map provided

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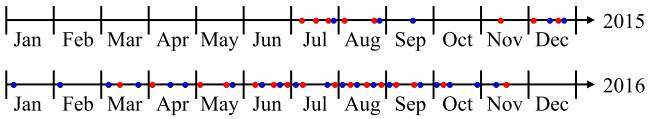


Fig. 6. Cloud-free images' timelines for 2015 and 2016 (red: orbit 051 and blue: orbit 094).

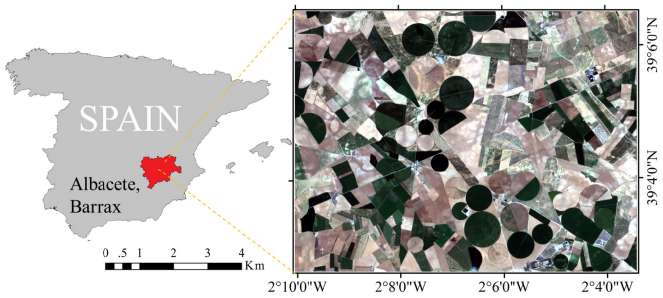


Fig. 7. Study area location in Barrax, Spain (RGB image on April 14, 2016).

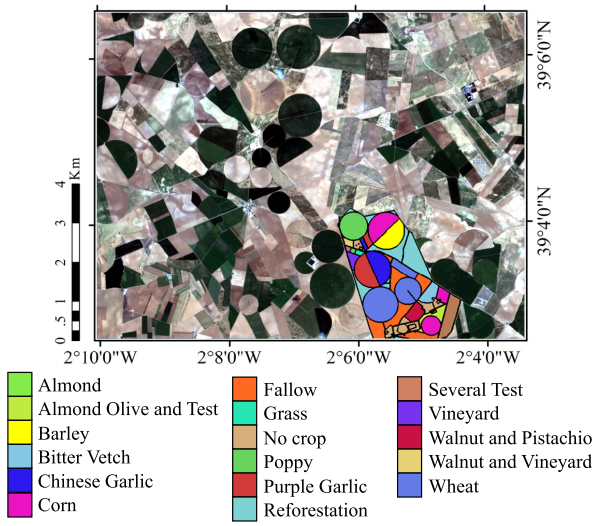


Fig. 8. Land cover map of 2015–2016 falling inside the study area [61].

by sen2cor. The incidence of irregularly distributed clouds may be reduced by focusing on the area of interest if it is smaller than an S2 tile [58]. Accordingly, each image in the S2-SITS was cut to the area of interest (726 × 954 pixels). However, the method effectively works on an entire tile as well. Finally, images were coregistered to mitigate the impact of misregistration on the next steps. The earliest image in the time series was selected as reference one, and all the others behaved as target images. Here, we used the area-based least-squares matching approach that provides subpixel precision suitable for the S2 spatial resolution and the task of crop fields analysis [59]. However, any SoA method can be employed. In the Barrax area, a total of 49 images were found to be cloud-free selected according to the sen2cor cloud mask. The temporal distribution of the free cloud images is shown in Fig. 6, where red dots represent the images acquired on orbit 051 and blue dots the ones on orbit 094. By considering the two orbits, we can see that there is at least one free cloud image per month (except for October 2015), but the sampling is irregular. The reference image to perform coregistration was acquired in the orbit 051.

The area around Barrax (see Fig. 7) has been used intensively for research on agriculture since many years given: 1) the different rotation practices; 2) the presence of crops such as the Alfalfa which is harvested several times per year because of its fast growing rate; and 3) the presence of the same type of crop under artificial irrigation and rainfed conditions [60]. Barrax area is an arid terrain with low annual precipitation and requires artificial irrigation. Fig. 7 shows the considered study area with a true color composition of an S2A image acquired on April 14, 2016.

In order to perform quantitative and qualitative validation, two kinds of reference information are used: 1) a land cover map built by considering acquisitions from a whole year and 2) vegetation/cropping cycles information collected by a survey filled by experts. At the time of writing, the most updated land cover map was available for the period 2015–2016 and covers a small area (see Fig. 8). This land cover map is

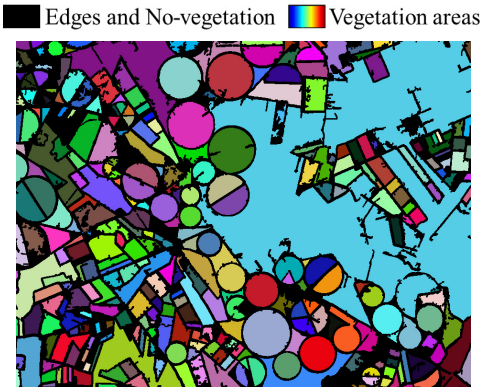


Fig. 9. Multitemporal vegetation map provided by the proposed method. Vegetation areas are colored randomly to illustrate that they are separate objects.

built considering the information reported by farmers at a given date of the year. The original mean NDVI-SITS, for each crop field in the area, was provided to 80 experts who were asked to identify the number of cropping cycles. In order to guarantee the reliability of the results, a total of 300 crop fields were considered and randomly split among 20 surveys, each containing 15 crop fields. For each crop field, an average of five–seven answers were collected and organized according to the majority voting. Land cover and cropping cycle information is considered as reliable, since they have been validated by experts that work in the study area and/or in the remote sensing field.

B. Multitemporal Vegetation Map

Once the S2A-SITS has been defined and preprocessed, we created an NDVI-SITS. To limit over splitting of fields, the NDVI-SITS is kept shorter than 1.5 years. This period guarantees a reliable result in Barrax, but it may vary across geographical areas. To compute $X_{\text{Spectral}_{\text{mask}}}$, the value of T in (2) was set to 0.1. The computation of $X_{\text{Edges}_{\text{mask}}}$ was conducted by several edge detection methods (i.e., Canny,

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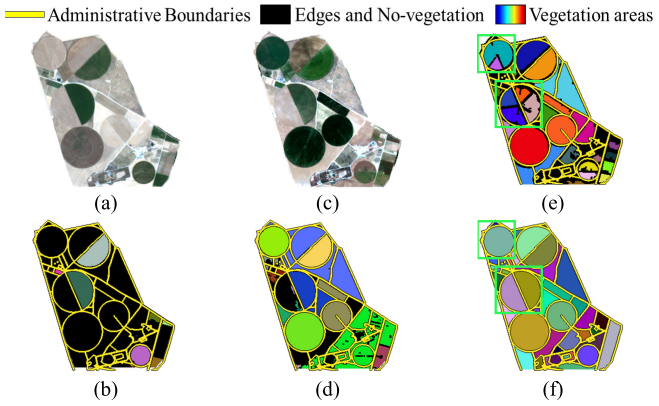


Fig. 10. (a) and (b) RGB and vegetation map of X_1. (c) and (d) RGB and vegetation map of X_{26}. (e) $X_{\text{veg}_{\text{map}}}$. (f) 2015–2016 reference map.

Prewitt, Sobel, Roberts, Laplacian, and Gaussian). Among them, the Sobel filter provided the best performance in terms of accuracy and computational time [43]. $X_{\text{Edges}_{\text{mask}}}$ was computed as indicated by (3). Equation (4) was applied to compute X_{mask}. $X_{\text{veg}_{\text{map}}}$ was obtained by an eight-connected-component labeling [44]. The tuning of the abovementioned parameters is easy and fast. The proposed approach resulted in 307 objects, as shown in Fig. 9. Black corresponds to no-vegetation and edges, and the other colors represent the vegetated areas. The area with ground reference information in the bottom right of Fig. 8 was used for performance analysis. This information is used for qualitative performance analysis only, since it provides information about static administrative boundaries but does not account for the spatio-temporal dynamic behavior of crops (e.g., rotation practices) that results in fragmented use of cadastral objects.

1) Comparison of Single-Date Versus Multiple-Date Vegetation Map Performance: The multitemporal vegetation map obtained with the proposed method is compared to a single-date vegetation map. The administrative boundaries in Fig. 8 are used as visual reference information. Fig. 10(f) shows the 2015–2016 reference map (closer look of the cadastral map shown in Fig. 8), and Fig. 10(e) shows $X_{\text{veg}_{\text{map}}}$. Single-date vegetation maps were extracted at t_1 and t_{26} by means of our proposed spectral and spatial fusion step [see Fig. 10(b) and (d)]. In the single-date option, underestimation occurs; by using t_1, only a few fields are detected because many were in their fallow season, while using t_{45}, many fields are merged (e.g., light green and blue ones), and similar situations can be seen for other images in the NDVI-SITS. Instead, the proposed spatio-temporal fusion approach accurately detects and separates crop fields. Fig. 9 shows the qualitative impression of this assertion. Another case is the one depicted in the light-green boxes in Fig. 10(e) where the administrative boundary encloses more than one crop type. This is because the administrative boundaries do not account for possible fragmentation (due to rotation practices), whereas the proposed method does. This example clarifies that crop field analysis by cadastral units may result in poor results. Small parcels falling in the area between the two green squares and corresponding to vineyards (and other tree-like crops, e.g., orchards) show a different behavior. In those cases, both the proposed algorithm and the single date maps fail to completely detect the fields. This is due to both the phenological behaviors and the tree-spatial distribution. In fact, trees are planted in rows that show large percentage of bare soil in between them. Thus, NDVI tends to be lower than crops showing more dense vegetation coverage. Such types of crops should be treated under different assumptions that will be further studied in the future works. Nevertheless, we expect the proposed method to effectively detect them after tuning parameters.

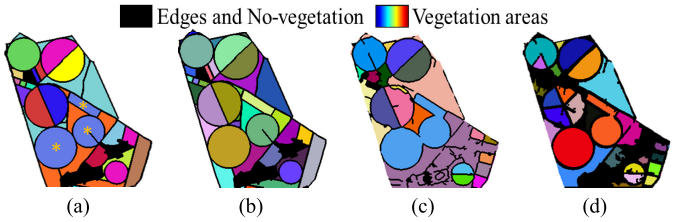


Fig. 11. (a) Land use map of 2015–2016. (b) Cadastral parcels and vegetation map obtained by (c) SoA method and (d) proposed approach.

2) Qualitative Performance Analysis (Proposed Approach Versus SoA): The multitemporal vegetation map obtained with the proposed method was compared with the one obtained by the SoA method in [62]. This method was selected among others [23], [24], [26] because it considers the rotation practices and the interannual variability of the crop fields. Therefore, it is the most similar to the proposed one and thus qualifies for a fair comparison. The method was developed for Landsat (temporal composites of five different spectral indices) and tested over South America. For each composite, it applies a series of multidirectional convolution kernels to extract edge gradient magnitude and uses a multiscale contrast adaptive histogram equalization and an adaptive thresholding to separate objects. Cropland and noncropland are distinguished by applying a supervised predictive model. For further details, the reader is referred to [62]. Fig. 11 compares the crop field map obtained in [62] with the one by the proposed approach. The proposed approach accurately separates vegetation areas from no-vegetation ones. This is the case of the different black patches (labeled as "no crop" in Fig. 8), which are identified by the SoA method as vegetation and even confused as a single big purple area in Fig. 11(c). It better separates the adjacent crop fields with the same land use. This is the case of crop fields marked with the asterisk in Fig. 11(a), which shows the same land use but belongs to different cadastral parcels [see Fig. 11(b)] and better detects the areas corresponding to vegetation. This is the case of the brown area in the bottom right, which appears to have a uniform land use [see Fig. 11(a)], but, in fact, it is labeled as "several test" area with multiple test cultivations. As one can see, both methods identify four crop fields, but the SoA confuses one of them with many other classes [large purple area in Fig. 11(c)]. Finally, from the computational viewpoint, the proposed approach produces results in 1 s, rather than the 60 s required by the SoA method.

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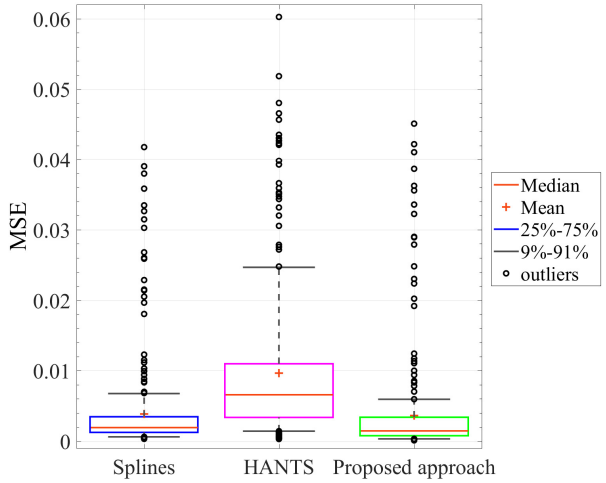


Fig. 12. Box plot of the MSE values for Splines, HANTS, and the proposed approach for each object in ω_v.

C. NDVI-SITS Reconstruction

The NDVI-SITS are reconstructed to continuous and regular acquisitions in time. Here, quantitative (based on MSE) and qualitative accuracy assessment is performed at single object level. To the best of our knowledge, methods do not exist in the literature, which are able to deal with both irregular and nonperiodic SITS. Therefore, the proposed approach was compared with methods, such as splines interpolation [63], Savistky–Golay, linear least squares, and double logistic function. Due to the space constraints, we provide a comparison with the two that performed better.

One of the considered methods is the Harmonic ANalysis of Time Series (HANTS) [64] and Splines. Though several other smoothing methods exist also able to work with irregular SITS, we selected HANTS over the others because it demonstrated to be the most effective in smoothing NDVI-SITS. Furthermore, the existing smoothing methods have been developed for low spatial resolution sensors (i.e., MODIS), a few of them have been tested in medium resolution sensors (i.e., Landsat), and none showed good performance for S2-SITS. HANTS is based on the Fourier series, and therefore, it assumes data harmonic behavior, *a priori* knowledge of the rotation practices (i.e., the number of cropping cycles), and regularly sampled data. In order to compare HANTS results with the ones of the proposed method, we modified the algorithm to deal with a near-daily acquisition. For nonintensive cultivations, the maximum expected number of cropping cycles is used to initialize HANTS algorithm, which is usually equal to 2. However, this value cannot be used for the considered test area since it is intensively cultivated. After some trials, the value was set to 4 as a tradeoff. It is worth recalling that in real case scenarios, such information is usually not available. Thus, we are favoring HANTS with respect to the proposed approach that does not require this kind of *a priori* knowledge. In the case of Spline interpolation, we expect to see a better performance with respect to HANTS, given that it does not rely on harmonic behaviors. At the same time, this penalizes the final smoothing result since Splines tends to overfit. In our experiments, the nonparametric regression parameter $T_{\mathrm{MSE}} = 0.01$ and $I = 100$ was set according to the literature [65], [66], and n_{HL_1} is set to 12, but it is worth noting that the approach resulted to be robust to this initialization by converging to a similar outcome for n_{HL_1} in the range $n_{\mathrm{HL}_1} = [1, 12]$. Fig. 12 shows the box plot of MSE values for all ω_v objects, computed only with respect to the mean NDV-SITS. Over the 307 objects in $X_{\mathrm{veg}_{\mathrm{map}}}$, the proposed approach achieved an average MSE value of 3.6×10^{-3}. Splines achieved an average MSE value of 3.9×10^{-3}, and this value is quiet close to that of the proposed approach since Splines tends to follow the real data. HANTS MSE average value is three-time larger (i.e., 8.7×10^{-3}). Both the proposed approach and the Splines interpolation show a lower MSE spread.

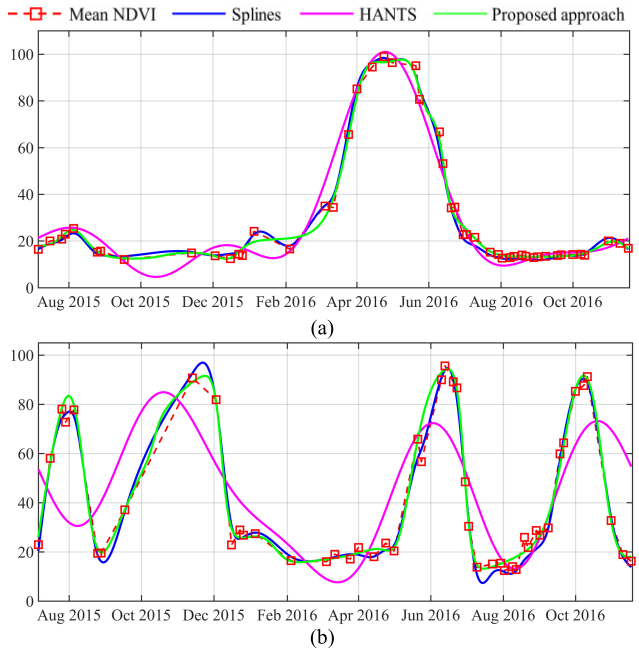


Fig. 13. Smoothed NDVI by Splines interpolation, HANTS, and proposed approach for (a) single-cropping cycle field and (b) four-cropping cycle field.

Fig. 13 shows the smoothed result from HANTS (magenta line), Splines (blue line), and the proposed approach (green line) for two of the objects in ω_v (NDVI values have been scaled by 100 for visualization purposes). It is possible to observe that the harmonic behavior of HANTS is successful with simple signatures showing only one cropping cycle [see Fig. 13 (a)], whereas it leads to poor estimation as the temporal signature becomes more complex [see Fig. 13(b) with four cropping cycles]. Furthermore, HANTS tends to underestimate the NDVI peak of each cropping cycle. This kind of behavior directly impacts on the accuracy of phenological parameters. Splines provides good performance in estimating the number of cropping cycles, but it tends to underestimate/overestimate the cropping cycle peaks and duration/shape. Instead, the proposed method produced an accurate result for both simple and complex cases while preserving the original shape and NDVI peaks. No significant differences have been recorded in the computational time, and no extra

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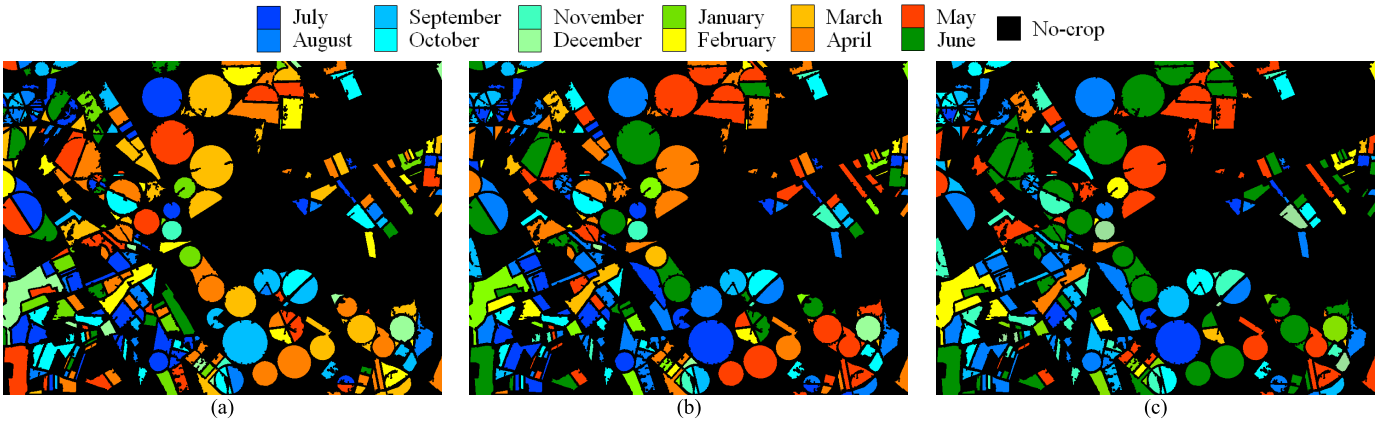


Fig. 14. Phenological parameter maps over 12 months (agrarian year) period for (a) BoS, (b) MoS, and (c) EoS. Crop fields with multiple phenological cycles being incomplete/truncated over the considered period may show EoS preceding BoS in the maps. This is only an apparent effect due to BoS and EoS belonging to two different cropping cycles.

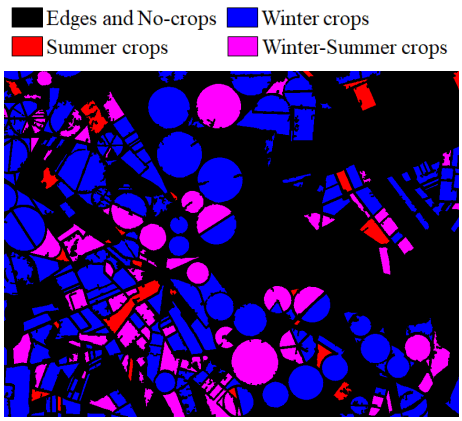


Fig. 15. Winter–Summer map for the Barrax area in the period from October 2015 to September 2016.

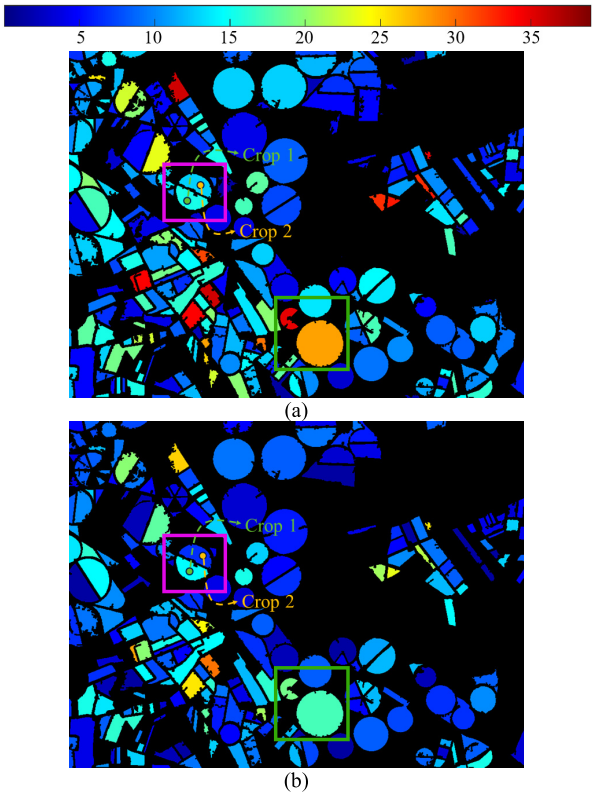


Fig. 16. Mean-cumulative (a) NDVI and (b) NDWI with $T = 0.5$.

computational complexity is added in any of the three algorithms while working on a regular temporal grid.

D. Spatio-temporal Information Extraction

Continuous and regular NDVI-SITS are the input for spatio-temporal information extraction. The goal of this step is to generate products that can support precision agriculture. Here, we analyze the products listed in Section II-C, but others can be defined. Without any loss of generality, the approach can be applied to features other than NDVI and used to generate other products. Quantitative analysis is carried out for the number of cropping cycles only, whereas qualitative analysis is carried out for the other products.

1) Quantitative Analysis (Cropping Cycles Extraction): The number of vegetation/cropping cycles was evaluated according to the information collected by the survey (see Section III-A for details). Each participant evaluated 60 objects, and each object was analyzed five–seven times. The outcomes of the survey and the accuracy analysis are provided in Table II. The proposed approach outperforms HANTS by about 6% and Spline by about 3%. For all the methods, the accuracy decreases as the temporal NDVI signature becomes more complex (i.e., the number of cropping cycles increases). However, the accuracy of HANTS and Splines methods decreases with a higher rate than that of the proposed approach. The lower accuracy of Splines mainly relates to overfitting, which results in phantom cropping cycles.

2) Qualitative Analysis (Phenological Parameters Maps): For the crop fields, the phenological parameter maps are built by color-coding in accordance with the time (day, week, or month) of phenological events. Examples for BoS, MoS, and EoS coded by month for a 12-month period (July 2015–June 2016) are shown in Fig. 14. The temporal information of these maps can be used to understand whether

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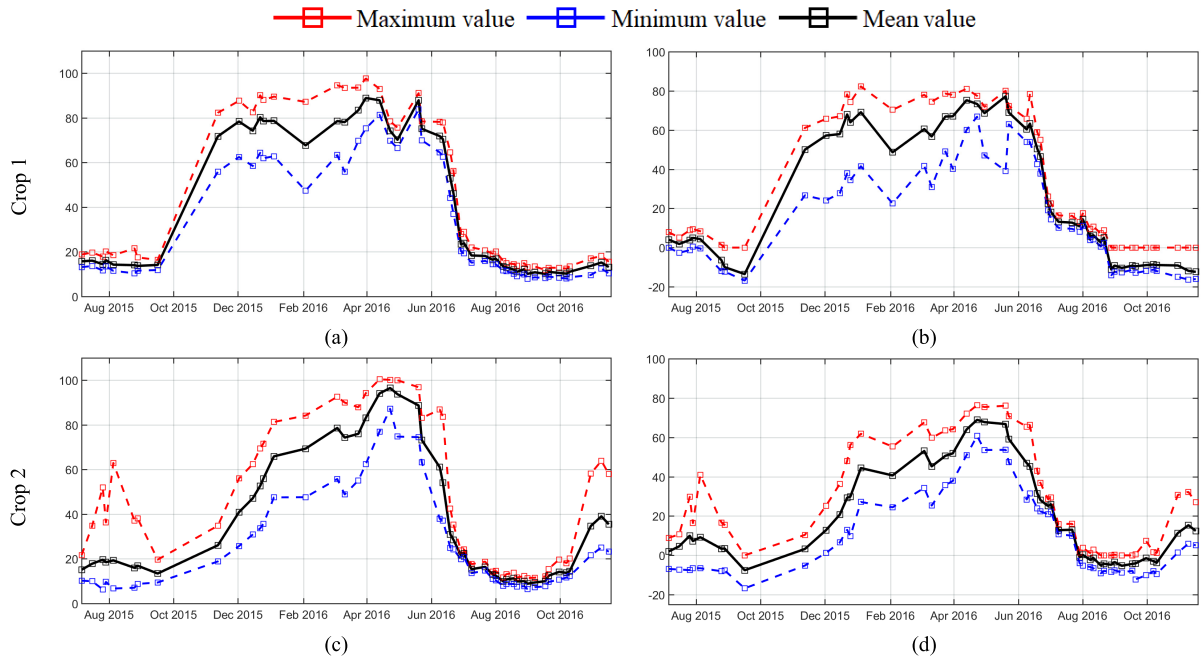


Fig. 17. (a) and (c) Statistical NDVI and (b) and (d) NDWI analysis for crop fields 1 and 2.

TABLE II

OVERALL ACCURACY IN THE DETECTION OF DIFFERENT CROPPING CYCLES FOR SPLINES INTERPOLATION, HANTS, AND THE PROPOSED APPROACH FOR DAILY TIME SERIES RECONSTRUCTION

# of cropping cycles	# of crop fields	Accuracy (%)		
		Splines	HANTS	Proposed Approach
0	54	100.00	100.00	100.00
1	107	94.27	93.59	95.54
2	93	89.56	87.65	92.06
3	19	86.08	82.72	89.01
4	8	84.34	80.07	87.54
5	2	83.75	79.51	86.93
6	3	82.87	78.67	86.01
7	5	81.44	77.32	84.54
Overall Accuracy		87.79	84.94	90.20

a crop has shifted phenological evolution. Comparison can be performed with respect to similar products from other years or from other crop fields. Another product based on phenological parameters is the winter/summer crop map (see Fig. 15). Winter crops are defined as the ones with BoS, MoS, and EoS in the period of October 2015 to March 2016. If the same metrics fall inside the period April–September 2016, the crops are considered as summer ones. Crop fields showing cycles in both periods are considered as winter and summer ones. The type of crop in the two periods might not be the same. Using the LoS parameter, we can distinguish between crop fields and other types of vegetation and build $X_{\text{crop}_{\text{map}}}$. For example, the large light blue area in Fig. 9 has a low NDVI value for most of the time because of sparse vegetation appearing and disappearing and is therefore labeled as ω_{ov}. If information regarding beginning and ending date of the agrarian year (and the specific rotation practices for the considered test site) are known, they can be used to compute maps in Fig. 15 over an appropriate time range to infer evidence about whether the type of crops has been watered artificially or not, and so on.

3) Qualitative Analysis (Cumulative Indices Maps): Spatio-temporal information extraction by means of a cumulative index (6) can be conducted for different radiometric indices in order to derive additional products. As an example, we extracted cumulative maps for NDVI and NDWI indices because of their relevance [36], [37], [67], [68]. Here, the accumulation is performed on thresholded indexes, with threshold $T = 0.5$ [36]). Fig. 16(a) and (b) shows the mean cumulative NDVI and NDWI, respectively; however, the pixel-based analysis can be conducted as well. Dark blue areas correspond to poor vegetation greenness/water content for every $t_n (n = 1, \ldots, 49)$, whereas blue and red shades correspond to the fields with a higher vegetation greenness/water content and thus more intensive use. Maps in Fig. 16 show spatially heterogeneous behaviors. A comparison between Fig. 16(a) and (b) shows that crop field with a similar cumulative NDVI does not correspond to a similar cumulative NDWI.

For example, crop fields in the magenta rectangle in Fig. 16 have a similar blue shade (they were vegetated for around 11 days), but they have a different mean cumulative NDWI. Other fields (green rectangle in Fig. 16) show similar shades for the NDWI (light blue), but different NDVI values. These kinds of outcomes can be used for analyses about irrigation practices, nutrients status, but also type of crops. A proper validation and interpretation would require ground truth and agricultural expert analysis. Let us take a closer look to the blue shade crop fields (crops 1 and 2) inside the magenta square by extracting some statistical information, such as minimum, maximum, and mean values for the NDVI [see Fig. 17(a) and (c)] and NDWI [see Fig. 17(b) and (d)] SITS. Despite similar mean-cumulative

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NDVI values, crops 1 and 2 show a different temporal evolution of NDVI and NDWI.

This is likely to depend on the difference of cultivation and irrigation requirements. Similar considerations can be conducted for all fields by taking advantage of the tool developed in this article and of experts for interpretation.

IV. Conclusion

A method to analyze the spatial and temporal evolution of small crop fields by jointly exploiting the spectral, spatial, and temporal information of S2-SITS has been presented. The proposed method includes three main steps: 1) spatio-temporal fusion; 2) daily time series reconstruction; and 3) spatio-temporal information extraction. The combination of these steps resulted in an effective tool that is suitable for the analysis of dense and intensive cultivated areas. Before applying each step, preprocessing is required, which renders the S2-SITS homogeneous and suitable for crop field identification and information extraction. Crop field identification relies on a spatio-temporal fusion of dense SITS that accounts for the dynamics of crops in space and time. The multitemporal crop field map can be easily updated as a new acquisition is available. Crop daily signature reconstruction was performed automatically by a nonparametric model that is able to deal with irregular samples data. Examples of spatio-temporal information/products that can be extracted from the resulting daily spectral signature at crop field level have been provided, including phenological parameters and cumulative radiometric index. However, without any loss of generality, other features can be considered depending on output products. Some of the novel contributions of this article were validated both in a quantitative and qualitative way, others only qualitatively. The validation demonstrated the effectiveness of the proposed approach and its relevance in precision agriculture.

As future developments, we consider the use of other radiometric indices, as well as the improvement of the spatio-temporal fusion step by means of other edge and shape detection methods. Additionally, a new set of rules needs to be introduced in order to deal with crops showing discontinuous vegetation coverage and high percentage of bare soil (i.e., vineyards and orchards). We also plan to expand the set of spatio-temporal products that can be generated and to improve the understanding of their impact on precision agriculture.

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